

Homework 2

There are a total of 6 questions.

Q1 (Reflection): Read the solution to HW 1.

- Are your own answers in line with the solutions? If not, list the questions you missed.
- Discuss what you could have done better.

Q2 (Conditional expectation): Let the random vector $(y, x)'$ have a normal distribution with mean vector $\mu = (\mu_y, \mu_x)'$ and covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_y^2 & \sigma_y \sigma_x \rho \\ \sigma_x \sigma_y \rho & \sigma_x^2 \end{pmatrix},$$

where σ_y and σ_x are the standard deviations and ρ is the correlation between y and x . The joint density is

$$f_{Y,X}(y, x) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{w} - \mu)' \Sigma^{-1}(\mathbf{w} - \mu)\right), \quad -\infty < y, x < \infty.$$

The determinant of the covariance matrix is

$$|\Sigma| = \sigma_y^2 \sigma_x^2 (1 - \rho^2),$$

and the inverse of the covariance matrix is

$$\Sigma^{-1} = \frac{1}{1 - \rho^2} \begin{pmatrix} \frac{1}{\sigma_y^2} & -\frac{\rho}{\sigma_y \sigma_x} \\ -\frac{\rho}{\sigma_y \sigma_x} & \frac{1}{\sigma_x^2} \end{pmatrix}.$$

Thus, the joint density can be written as

$$f_{Y,X}(y, x) = \frac{1}{2\pi\sigma_y\sigma_x(1 - \rho^2)^{1/2}} \exp\left(-\frac{1}{2(1 - \rho^2)} \left[\left(\frac{y - \mu_y}{\sigma_y}\right)^2 - 2\rho \left(\frac{y - \mu_y}{\sigma_y}\right) \left(\frac{x - \mu_x}{\sigma_x}\right) + \left(\frac{x - \mu_x}{\sigma_x}\right)^2 \right]\right).$$

The marginal density of x is

$$f_X(x) = \int_{-\infty}^{\infty} f_{Y,X}(y, x) dy = \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left(-\frac{1}{2} \left[\frac{x - \mu_x}{\sigma_x}\right]^2\right),$$

that is, normal with mean μ_x and variance σ_x^2 .

1. Derive the conditional distribution of y given x .

2. Compute the linear projection of y on $\mathbf{x} = (1, x)$. That is, derive and express $L(y|1, x)$ as a function of $\mu_x, \mu_y, \rho, \sigma_x, \sigma_y$.
3. Define $u = y - L(y|1, x)$. What is the distribution of u ?

Q3 (Linear projection): The textbook (Wooldridge)'s definition of the linear projection is slightly different from that was introduced in the lecture notes (Notes 01). Wooldridge defines the linear projection in the following way,

Define $\mathbf{x} = (x_1, \dots, x_K)$ as a $1 \times K$ vector, and make the assumption that the $K \times K$ variance matrix of $\mathbf{x}$ is nonsingular (positive definite). Then the linear projection of y on $1, x_1, x_2, \dots, x_K$ always exists and is unique:

$$L(y|1, x_1, \dots, x_K) = L(y|1, \mathbf{x}) = \beta_0 + \beta_1 x_1 + \dots + \beta_K x_K = \beta_0 + \mathbf{x}\beta,$$

where, by definition,

$$\beta = [Var(\mathbf{x})]^{-1}Cov(\mathbf{x}, y)$$

$$\beta_0 = E[y] - E[x]\beta = E[y] - \beta_1 E[x_1] - \dots - \beta_K E[x_K].$$

Explain why this definition coincides with the definition that is introduced in the lecture notes (Notes 01). Provide a formal derivation as well. Hint: Answer is in Notes 01.

Q4 (Asymptotics, asymptotic normality): Let $y_i, i = 1, 2, \dots$ be an independent, identically distributed sequence with $E[y_i^2] < \infty$. Let $\mu = E[y_i]$ and $\sigma^2 = Var(y_i)$.

1. Let $\bar{y}_N$ denote the sample average based on a sample size of N . Find $Var(\sqrt{N}(\bar{y}_N - \mu))$.
2. What is the asymptotic variance of $\sqrt{N}(\bar{y}_N - \mu)$?
3. What is the asymptotic variance of $\bar{y}_N$? Compare this with $Var(\bar{y}_N)$.
4. What is the asymptotic standard deviation of $\bar{y}_N$?

Q5 (Simulation and Asymptotics): Let's revisit [labnotes01.pdf](#). Redo the exercise with the following data generating process,

$$y_i = 1 + x_i + \varepsilon_i$$

where $\varepsilon_i = v_i - 20$ where $v_i \sim_{i.i.d} \chi_{20}^2$. Note that $E[\varepsilon] = 0$ and $Var(\varepsilon_i) = 40$. And, $x_i \sim_{i.i.d} N(1, 4)$. Note x_i and ε are assumed to be independent.

1. What is $Var(y_i)$?

2. Using the derivation from Q4, calculate the value of the asymptotic variance of $\bar{y}_N$ where $N = 100$.
3. Now, using Monte Carlo simulation as in the lab session note, calculate $Var(\bar{y}_N)$. Confirm that this number is similar to what you obtained from 2.
4. Compute skewness of $\bar{y}_N$ using Monte Carlo simulation for $N = 10, 100$, and 1000 (three different experiments). What do you see and explain your results.

Q6 (Paper question): Find a paper that uses the delta method in your field. If you can't find one, then find such a paper from the "American Economic Review", which is one of the premier journals in economics.

1. Find an academic paper¹ that (1) was published in one of the journals from your field of interest, AND (2) contains the word **delta method**, AND (3) the term **delta method** in the paper refers to the method that we learnt from the class, AND (4) applies the **delta method**.

One way to find such a paper is to use Google Scholar. Type the following in the search box

`source:"[name of the journal]" "delta method"`

2. Properly cite the paper you found (name of the author, the title of the article, year of publication, the name of the journal, etc.)
3. Read the paper and explain what is the main research question of the paper in one paragraph.
4. What is the parameter of interest in their empirical model, and why do the authors use the delta method?
5. (Optional reading; will not be graded). Read the following paper

Ver Hoef, J.M., 2012. Who invented the delta method?. *The American Statistician*, 66(2), pp.124-127.

(I put the copy of the paper in the HW section).

¹If you can't find such a paper in the field of your interest, then find one from the "American Economic Review", which is one of the premier journals in economics.